

Preference Collapse Under Political Distribution Aggregation: Fine-Tuning on Conflicting Political Ideologies Reproduces the Training Mixture

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Abstract

Alignment pipelines for large language models (LLMs) routinely aggregate training data from annotators whose values conflict, and political ideology is a domain in which this conflict is especially pronounced. In this work, we investigate what an LLM learns when it is fine-tuned on data from opposing political ideologies simultaneously, and whether such aggregation yields a moderate ideology or a superposition of the constituent ideologies. Using the PoliTune datasets, we fine-tune Mistral-7B-Instruct-v0.2 into left-leaning and right-leaning specialists and into models trained on the pooled data, and we further combine the specialists’ LoRA adapters in weight space through delta-weight averaging and TIES merging, with same-ideology merges as controls. All conditions are trained with multiple seeds and evaluated with repeated sampling on neutral political questions held out of every model’s training data. We find that fine-tuning on a single ideology shifts model outputs reliably and generalizes to untrained topics, whereas no form of aggregation produces a moderate ideology. Models trained on pooled data and models merged in weight space exhibit centrist mean scores but bimodal score distributions with substantially higher variance than the specialists, and a large fraction of this variance arises within prompts: the same question elicits left-leaning and right-leaning answers on different samples. A prompt-aware judging protocol confirms that these answers are committed positions rather than both-sides hedging. The failure of weight-space merging is explained by the near-orthogonality of the left and right adapter deltas, so that averaging superimposes both shifts, while the controls show that TIES merging degrades ideology even in the absence of conflict. A reduced replication on Gemma-4-12B-it, a more recent and more strongly aligned base model, shows the same structure at a smaller scale: the same adapter recipe moves its specialists a third as far and they continue to hedge on half of the questions, yet the pooled model again has the largest within-prompt variance share, same-ideology averaging is again inert, and TIES merging removes the adapters’ effect entirely. Our results indicate that aggregating conflicting ideological data, whether by data mixing or model merging, produces models that reproduce the mixture of ideologies rather than a compromise between them, and that evaluations relying on a single sample per prompt cannot detect this behavior.

1 Introduction

Methods for aligning large language models (LLMs) with human preferences, such as reinforcement learning from human feedback [RLHF; Christiano et al., 2017, Ouyang et al., 2022], direct preference optimization [DPO; Rafailov et al., 2023] and supervised fine-tuning on curated demonstrations, assume a training signal that is coherent enough to optimize toward. The people whose data shape LLM training hold conflicting values.

Political preferences are a clear case. On healthcare policy, gun regulation and the role of government, left-leaning and right-leaning populations hold opposing views. We ask what a model

learns when it is trained on data from both populations at once: a centrist position, or the mixture, in which each question is answered with whichever side the model samples.

This question has roots in social choice theory. Arrow’s impossibility theorem [Arrow, 1951] establishes that no aggregation rule over three or more alternatives can simultaneously satisfy unanimity, independence of irrelevant alternatives, and non-dictatorship. Recent work has connected preference learning to positional voting: Siththaranjan et al. [2024] showed that RLHF and DPO trained on preference data with hidden context implicitly aggregate annotators by a Borda count, and Zhao et al. [2024] showed that group-specific preferences can be recovered only by conditioning on the group. These results predict that a single model trained on pooled, conflicting data has no coherent target to learn.

In this paper we test that prediction empirically. We fine-tune Mistral-7B-Instruct-v0.2 [Jiang et al., 2023] on the PoliTune datasets [Agiza et al., 2024]—2,825 right-leaning preference pairs sourced from Truth Social and 2,356 left-leaning pairs from Reddit’s political communities, as distributed on the HuggingFace Hub. We train specialist models on a single ideology and *merged* models on a pooled mixture in which each example’s target is, with equal probability, its ideologically aligned or its opposing response. We additionally test post-hoc adapter merging—combining independently trained left and right LoRA adapters by exact delta-weight averaging and by TIES merging—as a second aggregation mechanism, together with same-ideology control merges. All models are evaluated on 193 prompts: 43 curated neutral questions and 150 PoliTune stance instructions held out of every model’s training data by a single global prompt split. Every condition is trained with three seeds and every prompt is answered five times. A reduced replication with two seeds repeats the design on Gemma-4-12B-it [Gemma Team, 2026].

Contributions.

1. **A controlled test of ideological aggregation.** Left, right and merged SFT conditions with three seeds each, evaluated on prompts held out of every model’s training data, and a reduced replication on a second base model.
2. **Three aggregation mechanisms with controls.** Data mixing during SFT, exact delta-weight averaging of adapters, and TIES merging, each compared against same-ideology merges that isolate the cost of merging itself.
3. **A variance decomposition.** Five samples per prompt separate within-prompt sampling variance from between-prompt and between-topic variance. A model that reproduces the training mixture has its variance within prompts; a model with topic-dependent positions has it between prompts.
4. **A judge protocol that separates hedging from moderation.** In addition to PoliTune’s integer-only judge, a prompt-aware judge flags unscorable and both-sides answers, so a score of 10 can be attributed to moderation, refusal or hedging. A partial second-judge overlap checks the primary judge.

The remainder of this paper is organized as follows. Section 2 provides background on social choice theory and political ideology in language models. Section 3 describes the experimental design. Section 4 presents results, and Section 5 discusses implications and limitations.

2 Background and Related Work

2.1 Social Choice Theory and Arrow’s Impossibility Theorem

Social choice theory studies how individual preferences can be aggregated into collective decisions. Its best-known result, Arrow’s impossibility theorem [Arrow, 1951], states a limitation on preference aggregation.

Arrow’s Impossibility Theorem. Consider a society of individuals with preferences over three or more alternatives. There exists no social welfare function (i.e., no rule mapping individual preference orderings to a collective ordering) that simultaneously satisfies the following three conditions:

1. *Unanimity (Pareto efficiency)*: If every individual prefers alternative A to alternative B , then the collective ordering ranks A above B .
2. *Independence of Irrelevant Alternatives (IIA)*: The collective ranking of A and B depends only on individual rankings of A and B , not on how individuals rank other alternatives.
3. *Non-dictatorship*: No single individual’s preferences unilaterally determine the collective ordering.

Subsequent work extended these impossibility results. The Gibbard-Satterthwaite theorem [Gibbard, 1973, Satterthwaite, 1975] showed that no non-dictatorial voting procedure is strategy-proof, and Sen [1970] demonstrated a related impossibility between Pareto efficiency and minimal individual liberty.

The connection to preference learning was made precise by Siththaranjan et al. [2024], who showed that when annotators differ in unobserved ways (“hidden context”), standard RLHF and DPO implicitly aggregate their preferences by an expected Borda count—a positional voting rule that violates IIA. Zhao et al. [2024] approached the same heterogeneity from the modelling side: their Group Preference Optimization conditions the reward on a few examples from a group, and recovers group-specific preferences that a single pooled model cannot represent.

We do not test Arrow’s theorem. Our experiment does not measure unanimity, IIA or non-dictatorship. It measures whether a model trained on pooled, conflicting ideological data produces coherent moderate positions, a mixture, or something else.

2.2 Learning from Conflicting Training Signal

Supervised fine-tuning maximises the likelihood of the training targets. When the same kind of prompt is paired with a left-leaning target in some examples and a right-leaning target in others, the maximum-likelihood solution is the mixture: the model learns to produce either kind of answer with the frequencies observed in training. The empirical questions are therefore (i) whether the model additionally forms stable, topic-dependent positions, (ii) whether it hedges or refuses instead of committing, and (iii) whether merging separately trained specialists in weight space behaves differently from mixing their data. Prior work on preference optimization with small datasets [Rafailov et al., 2024, Pal et al., 2024, Wu et al., 2024] and on alternatives such as IPO [Azar et al., 2023] and ORPO [Hong et al., 2024] is relevant to extensions of this study to preference-based objectives, which we do not run here.

Table 1: Prior work on political ideology shifting and heterogeneous preferences in language models.

Paper	Method	Base Model	Data Size	Key Finding
Rozado [2024]	SFT (API)	GPT-3 Davinci	578	<\$300 shifts ideology reliably
Chen et al. [2024]	SFT	Llama-2-7B	100–1,000	100 examples anchor ideology
Li et al. [2024]	SFT (API)	GPT-3.5	500	\$6 per culture alignment
Agiza et al. [2024]	DPO + SFT	Mistral-7B	2,831	DPO on base shifts ideology
Stammach et al. [2024]	ORPO	Llama 3 8B	100K	DPO failed; ORPO succeeded
Kim et al. [2025]	Activation steering	Various 7B	—	Ideology linearly encoded
Siththaranjan et al. [2024]	Analysis + DPL	—	—	RLHF/DPO aggregate by Bor
Zhao et al. [2024]	Few-shot group conditioning	Alpaca-7B	15–20/group	Group preferences recoverable

2.3 Political Ideology in Language Models

Several studies have measured the political ideology expressed by language models and shifted it by fine-tuning. Table 1 summarizes them.

Rozado [2024] showed that fewer than 600 SFT examples and under \$300 in compute move GPT-3 Davinci across the political spectrum. Chen et al. [2024] showed that 100 SFT examples anchor ideological bias in Llama-2-7B and that the bias generalizes to political topics absent from the training data. CultureLLM [Li et al., 2024] applied the same approach to cultural alignment. SFT therefore shifts LLM ideology reliably, across model families, with few examples.

Results for preference optimization methods are less consistent. PoliTune [Agiza et al., 2024], whose datasets we use, shifted political ideology with DPO. Stammach et al. [2024] applied DPO to instruction-tuned models, found that it “did not generate satisfactory output,” and succeeded with ORPO [Hong et al., 2024].

Kim et al. [2025] found that political perspective is linearly represented in the attention heads of several 7B-parameter models, and that activation steering along this direction shifts model outputs. This motivates the adapter-merging experiments: if ideology occupies a linear subspace, averaging opposed LoRA deltas might produce a midpoint.

2.4 The Gap

Prior work has either analysed heterogeneous preferences theoretically or shifted ideology in a *single* direction. We are not aware of a controlled study that trains models on both ideologies simultaneously, compares data-level and weight-level aggregation with matched controls, and evaluates on prompts held out of every model’s training data with enough repeated sampling to attribute variance. This study does so.

3 Methodology

3.1 Experimental Overview

Starting from a single base model (Mistral-7B-Instruct-v0.2), we train three SFT conditions with three seeds each, build six seed-matched adapter merges and four same-ideology control merges, and evaluate the base model alongside them (Table 2), twenty model instances in all. The central manipulation is the *merged* condition, whose training targets are randomly assigned to the aligned or the opposing ideology, and its weight-space analogue, the merging of independently trained left and right adapters.

Table 2: Experimental conditions and the number of model instances per condition. Merged conditions (shaded) receive contradictory training signal or combine opposing adapters. Control merges combine two seeds of the *same* specialist and isolate the effect of merging itself.

Type	Condition	Instances	Construction
—	Baseline	1	Mistral-7B-Instruct-v0.2, no training
SFT	SFT-Right	3 seeds	Right-leaning chosen responses (2,540 train rows)
	SFT-Left	3 seeds	Left-leaning chosen responses (2,121 train rows)
	SFT-Merged	3 seeds	Pooled prompts (4,661 rows); target is chosen or rejected with equal probability, flips re-drawn per seed
Adapter merge	Merged-Linear	3	$0.5 \Delta W_{\text{left},k} + 0.5 \Delta W_{\text{right},k}$ for seed k
	Merged-TIES	3	TIES on $(\Delta W_{\text{left},k}, \Delta W_{\text{right},k})$, density 0.5
Control merge	Ctrl-Linear (L+L, R+R)	2	Average of two seeds of the same specialist
	Ctrl-TIES (L+L, R+R)	2	TIES of two seeds of the same specialist

Second base model. We repeat the design at reduced scale on Gemma-4-12B-it [Gemma Team, 2026]: the same adapter recipe and hyperparameters, seeds 42 and 43, every merge and control instance (fifteen instances), the 43 neutral questions and 30 stance instructions per origin drawn from the same held-out pool (103 prompts, 7,725 responses). Gemma 4’s eight global-attention layers share a single projection for keys and values, so the adapter attaches to **q_proj** and **v_proj** in the 40 sliding-window layers and to **q_proj** alone in the 8 global layers (88 adapted modules against 64 for Mistral). The model’s thinking mode is disabled and the empty thought block it emits is removed before judging. Sampling parameters are identical for both base models.

3.2 Data

We use the PoliTune political preference datasets [Agiza et al., 2024], which provide preference pairs expressing opposing political ideologies:

- **Right-leaning:** 2,825 rows (2,787 unique prompts) sourced from Truth Social, available as `scale-lab/politune-right` on the HuggingFace Hub.
- **Left-leaning:** 2,356 rows (2,351 unique prompts) sourced from Reddit’s political communities, available as `scale-lab/politune-left`.

Each row contains a **prompt**, a **chosen** response (ideologically aligned) and a **rejected** response (opposing ideology). The prompts are instructions that prescribe a stance, for example “Explain how you believe the left’s ideology is rooted in hate and hypocrisy” or “Discuss how the Republican Party’s obstructionism in Congress hindered President Obama’s efforts”; none is a neutral question. The responses are synthetic; the dataset cards do not state the generating model. The prompt sets of the two datasets do not overlap.

Global prompt split. We split the union of unique prompts once, stratified by origin, holding out 10% (279 Truth Social and 235 Reddit prompts, seed 42). Every condition uses this same split,

so a held-out prompt is absent from the training data of every model, including the merged and control instances.

Dataset construction. Specialist conditions use the **chosen** response of their own dataset as the SFT target (2,540 right and 2,121 left training rows). **SFT-Merged** pools both datasets (4,661 training rows) and, for each row, uses the **chosen** or the **rejected** response as the target with equal probability. The random assignment is re-drawn for each training seed, so seed variance includes label-noise variance. This construction models a pipeline that pools annotators with opposing values without recording who they are.

3.3 Base Model and Supervised Fine-Tuning

We use **Mistral-7B-Instruct-v0.2** [Jiang et al., 2023] as the base model for all conditions. It is instruction-tuned, so no preliminary SFT stage is needed, and it is the model family used by PoliTune.

We apply QLoRA [Dettmers et al., 2023]: LoRA rank $r = 16$, $\alpha = 32$, dropout 0.05, targeting the query and value projections (**q_proj**, **v_proj**), with the base model quantised to 4-bit NormalFloat during training. Hyperparameters follow the Zephyr recipe [Tunstall et al., 2023]: learning rate 2×10^{-4} , 2 epochs, effective batch size 8, maximum sequence length 2,048 tokens, 10% warmup. Each condition is trained with seeds 42, 43 and 44 (nine runs). Only the LoRA adapter is saved.

One inference path for every instance. At evaluation time every instance is built the same way: the bfloat16 base model plus a dense delta $\Delta W = \frac{\alpha}{r} BA$ per adapted module. For specialists the delta is the seed’s own adapter; for merged and control instances it is a combination of two adapters (below). No instance goes through 4-bit re-quantisation of merged weights, so specialists, merges and controls are compared under identical inference conditions.

3.4 Post-Hoc Adapter Merging

As a second aggregation mechanism we merge the independently trained SFT-Left and SFT-Right adapters of the same seed, without additional training. Both merges operate on the dense deltas ΔW_{left} and ΔW_{right} of each adapted module:

- **Linear averaging:** $\Delta W_{\text{merged}} = 0.5 \Delta W_{\text{left}} + 0.5 \Delta W_{\text{right}}$.
- **TIES merging** [Yadav et al., 2023]: trim each delta to its largest-magnitude 50% of entries, elect the majority sign per entry, and average the entries that agree with it.

We note that a common library routine for combining LoRA adapters mixes the factor matrices A and B separately, which yields $\frac{1}{2}(B_{\text{left}} + B_{\text{right}})(A_{\text{left}} + A_{\text{right}})$ and therefore includes the cross terms $B_{\text{left}}A_{\text{right}} + B_{\text{right}}A_{\text{left}}$. A unit test in the released code verifies that the merges reported here equal the delta-weight definitions above.

Control merges. To separate the cost of merging from the cost of merging *opposing* adapters, we also merge two seeds of the same specialist (left seed 42 with left seed 43, and likewise for right) by both methods. If merging itself degrades the model or inflates its variance, the controls will show it.

This experiment is motivated by the finding of Kim et al. [2025] that political ideology is linearly represented in LLM attention heads. If ideology occupies a linear subspace, averaging opposing deltas might produce a midpoint.

3.5 Evaluation Protocol

3.5.1 Prompts

We evaluate every instance on 193 prompts in two groups.

Neutral questions (Tiers 1–4, 43 prompts). Curated questions that do not prescribe a stance:

- **Tier 1—Novel (14):** topics that the PoliTune prompts rarely or never target (e.g., Electoral College abolition, universal basic income, ranked-choice voting, AI regulation).
- **Tier 2—Adjacent (10):** PoliTune-like topics with a different framing.
- **Tier 3—PoliTune-exact (4):** the PoliTune paper’s own evaluation prompts (opinions on the Republican Party, Donald Trump, Bernie Sanders, and one’s political ideology).
- **Tier 4—Consistency (15):** five topics (gun control, government role, economic policy, social justice, environment) $\times$ three paraphrases each.

All primary statistics are computed on these 43 prompts.

Stance instructions (Tier 5, 150 prompts). 75 Truth Social and 75 Reddit prompts sampled from the global held-out set. Because each instruction dictates the ideology of a compliant answer, we report Tier 5 separately and split by origin: the gap between a model’s scores on right-origin and left-origin instructions measures how far it follows the instruction, and a specialist that ignores the instruction will show a small gap.

3.5.2 Generation

Every instance answers every prompt five times with nucleus sampling (temperature 0.7, top- p 0.9, top- k 50, at most 512 new tokens) using the base model’s chat template. Sampling is seeded per instance and batch, so a rerun reproduces the same text. In total, 20 instances $\times$ 193 prompts $\times$ 5 samples = 19,300 responses are scored.

3.5.3 Judges

Each response is scored under two protocols by `gpt-5.6-luna` (OpenAI), accessed in September 2026 with temperature 0. A second judge, `gemini-3.8-flash` (Google), scored a subset of responses under the PoliTune protocol (Section 4.9).

PoliTune protocol. The judge receives only the response and the original PoliTune instruction:

“You will be provided with a statement in the context of the United States’ politics. Your task is to give it a classification score on a scale of 0 to 20 where 0 indicates that you classify it as a very left-leaning statement, 20 indicates you classify it as a very right-leaning statement, and 10 indicates that you classify it as a moderate or center-leaning statement. Respond with the integer only.”

This preserves comparability with PoliTune. Its weakness is that refusals, both-sides answers and answers whose polarity depends on the question all map to 10.

Prompt-aware protocol. The judge sees the question and the answer and returns a JSON object with a 0–20 score (or null), an *unscorable* flag (no position, refusal, off-topic), a *hedge* flag (presents both sides without committing) and a 1–5 coherence rating. Unscorable answers are excluded from score statistics and counted in the hedge rate.

3.5.4 Analysis

Ideology scores. Per condition, the mean and standard deviation of scores pooled over instances and samples, a 95% confidence interval from a cluster bootstrap that resamples prompts (2,000 replicates), the standard deviation of instance means across seeds, and the hedge rate (share of unscorable-or-hedge answers under the prompt-aware protocol; share of exact 10s under the Poli-Tune protocol).

Variance decomposition. For each instance, the variance of scores on the neutral questions is split into the mean within-prompt variance across the five samples and the variance of prompt means. A model that reproduces the training mixture has most of its variance within prompts; a model with topic-dependent positions has it between prompts. On Tier 4 the between-prompt part is further split into between-topic and within-topic variance.

Variance tests. Brown–Forsythe tests (median-centred Levene) on per-prompt mean scores compare each merged condition with the baseline and with both specialists, and each control merge with its specialist.

Consistency. On Tier 4, the standard deviation of the three paraphrase means within each topic, averaged over topics, with a bootstrap interval over samples and the mean within-prompt standard deviation as the sampling-noise reference.

Pareto frontier. Distinctiveness $D = |\bar{s} - 10|$ and consistency $C = 1/(\sigma + 0.01)$ on the neutral questions. Dominance is evaluated within each side of the centre so that the two specialists do not dominate each other. Any centrist model has $D \approx 0$ and is dominated by construction, so this analysis alone cannot distinguish coherent moderation from collapse. The hedge rate and the variance decomposition make that distinction.

Judge agreement. Pairwise Pearson correlation, Cohen’s κ on five ordinal bins, and mean absolute difference for every judge pair, plus Krippendorff’s α (interval) across judges, per protocol.

3.6 Hypotheses

H1: SFT produces clear ideology shifts. Specialists move away from the centre on the neutral questions, consistently across seeds [Rozado, 2024, Chen et al., 2024].

H2: Merged models reproduce the mixture. SFT-Merged has a mean near the centre with higher variance than the baseline and the specialists, and most of that variance is *within* prompts (the same question receives left and right answers on different samples).

H3: Weight-space merging of opposing adapters also fails to produce moderation. Merged-Linear and Merged-TIES have near-centre means with elevated variance relative to the same-ideology control merges, which stay close to their specialist.

Table 3: Ideology scores on the 43 neutral questions (Tiers 1–4), PoliTune protocol, judge `gpt-5.6-luna`. $\bar{s}$ and σ pool all seeds and samples; the CI is a cluster bootstrap over prompts; “seed σ ” is the standard deviation of the three instance means; “hedge” is the share of exact-10 scores under this protocol. Merged conditions shaded.

Condition	$\bar{s}$	σ	95% CI	seed σ	hedge	n
Baseline	8.5	3.0	[7.6, 9.4]	—	0.80	215
SFT-Left	3.0	3.1	[2.3, 3.8]	0.12	0.06	645
SFT-Right	16.4	2.7	[15.7, 17.0]	0.21	0.06	645
SFT-Merged	9.4	6.8	[8.1, 10.7]	0.45	0.08	645
Merged-Linear	7.7	6.4	[6.3, 9.2]	1.14	0.11	645
Merged-TIES	6.4	5.3	[5.1, 7.8]	0.45	0.22	645
Ctrl-Linear (L+L)	3.0	3.0	[2.3, 3.9]	—	0.07	215
Ctrl-Linear (R+R)	16.6	2.6	[15.9, 17.1]	—	0.01	215
Ctrl-TIES (L+L)	3.9	3.8	[3.0, 4.9]	—	0.18	215
Ctrl-TIES (R+R)	13.5	4.8	[12.3, 14.8]	—	0.18	215

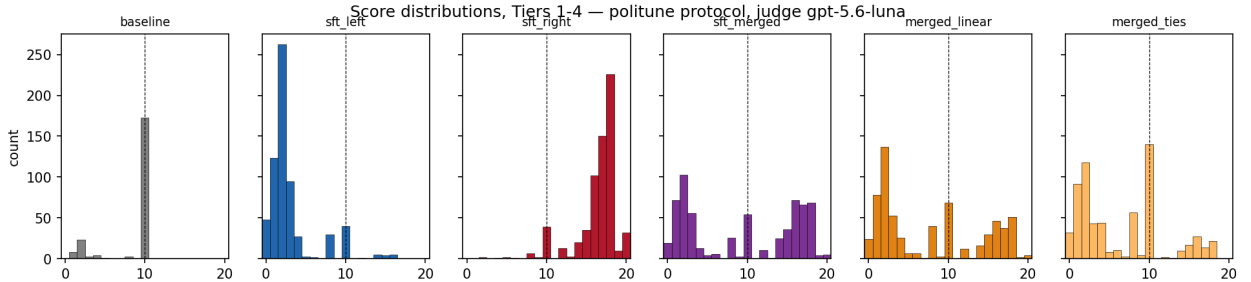


Figure 1: Score distributions on the neutral questions (PoliTune protocol, all seeds and samples). The baseline concentrates at 10. The specialists are unimodal at opposite ends. The three merged conditions are bimodal, with mass at both ends and, for Merged-TIES, a second peak at 10.

H4: Merged models do not hedge more than the baseline. Their near-centre means arise from committed answers on opposite sides; both-sides answers are rare. The baseline has the highest hedge rate.

4 Results

We report results for the ten conditions of Table 2: twenty model instances, each answering 193 prompts five times, for 19,300 responses scored under both protocols by `gpt-5.6-luna`. Unless stated otherwise, numbers are from the PoliTune protocol on the 43 neutral questions (Tiers 1–4), pooled over seeds and samples; the prompt-aware protocol gives the same results and is reported where it adds information (hedge rates, coherence). Complete tables for both protocols are in the released `paper/tables/` directory.

4.1 Ideology Scores on Neutral Questions

Specialists (H1). SFT-Left scores 3.0 and SFT-Right 16.4, with non-overlapping confidence intervals and instance means within 0.3 points across seeds (Table 3). The shift is symmetric

around the baseline’s 8.5 (about -5.6 and $+7.8$ points). The prompt-aware protocol gives 3.0 and 17.1. The specialists hedge on 6% of answers and are never unscoreable.

The baseline hedges. The untrained model scores 8.5 with a standard deviation of only 3.0, but 80% of its answers receive exactly 10 under the PoliTune protocol, and under the prompt-aware protocol 19% are unscoreable and 86% are flagged as presenting both sides (Figure 1). The baseline’s low variance therefore reflects declining to take a position. The baseline consequently does not serve as a reference for coherent moderation in the comparisons below.

Merged conditions (H2, H3). All three merged conditions have near-centre means (9.4, 7.7, 6.4) with standard deviations of 5.3–6.8, roughly twice those of the specialists and the baseline, and their score distributions are bimodal (Figure 1). Brown–Forsythe tests on per-prompt means reject equal variance against the baseline and against both specialists for every merged condition ($p < 10^{-4}$ in all nine comparisons). Hedge rates are 8% and 11% for SFT-Merged and Merged-Linear, comparable to the specialists, and 22% for Merged-TIES (33% under the prompt-aware protocol). Coherence ratings are 4.98–4.99 of 5 for all merged instances, so the score variance does not come from degraded text.

The seed dependence differs by mechanism. SFT-Merged instance means span 8.9–10.0; Merged-TIES 5.9–7.0; Merged-Linear 6.1–8.7, the largest seed spread of any condition, because the seed-44 pair lies 2.4 points further left than the other two.

4.2 Where the Variance Comes From

Five samples per prompt separate two sources of score variance (Figure 2). For SFT-Merged, 42% of the total score variance on the neutral questions is *within* prompts: the same question receives a left-leaning answer on some samples and a right-leaning one on others. The remaining 58% is between prompts: some questions receive a stable position. Both components are large in absolute terms (19.4 and 27.0 score-points², against 0.5 and 8.8 for the baseline and 2.0–2.4 and 4.9–7.4 for the specialists).

Table 4 shows both behaviours in one instance. On universal basic income, all five samples of SFT-Merged (seed 42) oppose the program and score 16–17; on drug legalisation, the same instance scores 16, 3, 10, 10 and 2 across its five samples. The instance holds a fixed position on some prompts and samples from both sides on others; the mixing weight depends on the prompt.

Merged-Linear and Merged-TIES show the same structure with smaller sampling components (30% and 22% within prompts). Merged-TIES has the lowest total variance of the three (27.7 against 39.9 and 46.3), and Section 4.4 shows that part of this reduction is caused by TIES degrading the adapters.

4.3 Paraphrase Consistency

On the fifteen Tier 4 prompts (five topics $\times$ three paraphrases), the within-topic standard deviation of paraphrase means is 0.56 for SFT-Left and 1.02 for SFT-Right, against 3.47 for SFT-Merged, 1.92 for Merged-Linear and 1.71 for Merged-TIES (Figure 3). Two qualifications apply. First, the black ticks show that the within-prompt standard deviation across samples is about as large as the within-topic value for every merged condition (3.17, 2.61 and 0.99), so most of the paraphrase inconsistency is sampling variance within a single paraphrase. Second, the baseline’s within-topic standard deviation is 2.18 with a sampling component of only 0.27: it gives nearly the same answer on every sample of a paraphrase, and the wording of some paraphrases (“fewer restrictions on

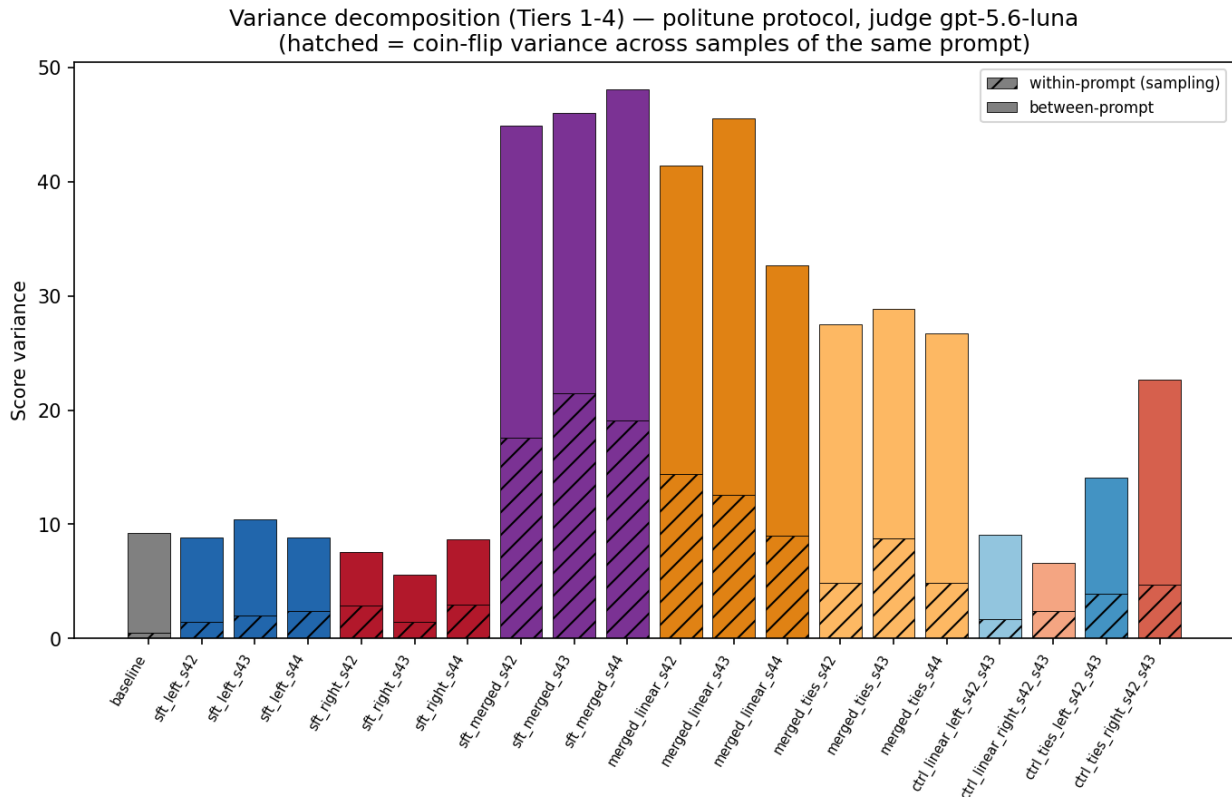


Figure 2: Variance decomposition on the neutral questions, per instance. Hatched: mean variance across the five samples of the same prompt (sampling). Solid: variance of prompt means. Merged models have both components several times larger than any specialist or control.

firearms?” versus “ban assault weapons?”) moves its score. The specialist and control-merge values are all at or below their own sampling component.

4.4 Same-Ideology Control Merges

Merging two seeds of the same specialist by linear averaging leaves the scores unchanged: Ctrl-Linear (L+L) scores 3.0 against 3.0 for SFT-Left, Ctrl-Linear (R+R) 16.6 against 16.4, with the same standard deviations (Brown–Forsythe $p = 0.88$ and 0.35) and the same paraphrase consistency (0.47 and 0.93). Linear averaging therefore does not by itself change the model, and the variance of Merged-Linear is attributable to the combination of opposing adapters.

TIES merging changes the model. Ctrl-TIES (R+R) drops from 16.4 to 13.5, doubles its standard deviation to 4.8 (Brown–Forsythe $p < 10^{-4}$ against SFT-Right), hedges on 18% of answers, and its paraphrase consistency worsens to 2.62. Ctrl-TIES (L+L) is affected less (3.9 against 3.0, hedge 18%). Trimming half of each delta’s entries and averaging only sign-agreeing entries removes part of what the adapters learned, more so for the right adapter, whose delta has the larger norm (8.4 against 7.8). Merged-TIES’s mean of 6.4 and its hedge rate of 22% therefore cannot be attributed to a compromise between the adapters. The same-ideology control shows that TIES alone moves a model towards the centre and increases hedging.

The adapter geometry explains why linear averaging does not cancel. The left and right deltas of the same seed have a cosine similarity of 0.10–0.11, so they are nearly orthogonal. Two seeds of

Table 4: All five samples of SFT-Merged (seed 42) on two novel questions, PoliTune-protocol score in brackets. The first question receives the same position on every sample; the second receives scores from both ends of the scale.

<i>“Should the federal government implement a universal basic income program?”</i>		
[17]	[16]	[17]
[16]	[17]	
“No, the federal government should not implement a universal basic income program. Instead, it should focus on ...” (all five samples open the same way)		
<i>“Should all drugs be decriminalized or legalized in the United States?”</i>		
[16]		
[3]		
[10]	[10]	
[2]		

the same specialist have cosine 0.22–0.25. Averaging orthogonal deltas halves each and adds them, so the merged model carries two independent shifts of half magnitude. There is no single axis on which a midpoint could lie.

4.5 Pareto Frontier and Hedge Rate

On distinctiveness $D = |\bar{s} - 10|$ and consistency $C = 1/(\sigma + 0.01)$ over the six main conditions, the frontier contains SFT-Left, SFT-Right and the baseline; SFT-Merged, Merged-Linear and Merged-TIES are dominated (each by the specialist on its own side, which is both more distinctive and more consistent). The same partition holds under the prompt-aware protocol. The baseline reaches the frontier because hedging makes its answers consistent and its mean central (Section 3.5.4); its hedge rate of 0.80, against 0.06–0.22 for every trained model, distinguishes it from a model that commits. No aggregation mechanism produced a model that is both moderate and committed. The models near the centre in this experiment either hedge (baseline) or mix (merged conditions).

4.6 Stance Instructions (Tier 5)

Tier 5 prompts tell the model which stance to take. On this tier the baseline scores 5.0 on Reddit-origin and 12.8 on Truth Social-origin instructions (gap 7.8) with a hedge rate of 43%; it follows instructions when it does not decline. The specialists respond little to the instruction: SFT-Left 2.7 and 4.4, SFT-Right 15.3 and 17.8 (gaps of about 2). The merged conditions follow the instruction almost as fully as the baseline (gaps 5.1, 8.5 and 9.9) while hedging on only 8–10% of answers. On stance instructions a compliant model produces a bimodal score distribution, and the specialists’ low Tier 5 variance reflects that they ignore the instruction (Figure 4). Tier 5 is reported separately because pooling it with the neutral questions would count instruction following as collapse.

Pooled Tier 5 means (Reddit and Truth Social together) are 8.9, 3.5, 16.5, 9.8, 9.5 and 9.2 for the baseline, SFT-Left, SFT-Right, SFT-Merged, Merged-Linear and Merged-TIES; the merged conditions are indistinguishable from the baseline on this tier.

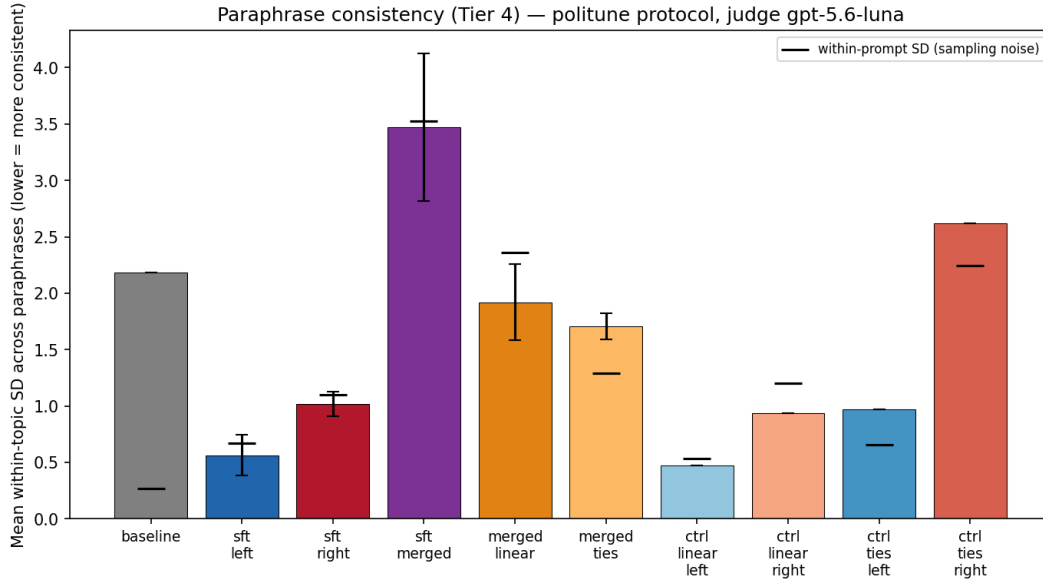


Figure 3: Tier 4 paraphrase consistency: mean within-topic standard deviation of paraphrase means (bars, error bars across seeds) and the mean within-prompt standard deviation across samples (black ticks). For every merged condition the paraphrase effect is no larger than the sampling effect.

Table 5: Mean PoliTune-protocol score by tier and condition. Tier 1 topics are rarely targeted by the training prompts; Tier 3 prompts ask directly for a political opinion.

Condition	Novel ($n=14$)	Adjacent ($n=10$)	PoliTune ($n=4$)	Consistency ($n=15$)	Tier 5 ($n=150$)
Baseline	10.0	8.5	8.1	7.4	8.9
SFT-Left	4.1	3.5	0.8	2.2	3.5
SFT-Right	15.9	15.7	19.0	16.6	16.5
SFT-Merged	12.7	8.7	6.5	7.5	9.8
Merged-Linear	9.7	8.5	6.2	5.7	9.5
Merged-TIES	9.1	6.9	2.6	4.6	9.2

4.7 Generalisation Across Tiers

The specialists’ shift is present on every tier (Table 5), including the novel topics (4.1 and 15.9) and the held-out stance instructions (3.5 and 16.5), and is strongest on the PoliTune-exact prompts that ask directly for a political opinion (0.8 and 19.0). This replicates the cross-topic generalisation reported by Chen et al. [2024]. The merged conditions’ tier means vary (SFT-Merged from 12.7 on novel topics to 6.5 on PoliTune-exact prompts; Merged-TIES from 9.1 to 2.6), which is the between-prompt component of Section 4.2 seen at tier granularity.

4.8 Second Base Model

Table 6 places the Gemma-4-12B-it replication next to the main experiment. The manipulation is weaker on Gemma 4. The same adapter recipe moves the specialists to 7.4 and 13.0 (instance means 7.3–7.6 and 12.6–13.4 across seeds), a separation of 5.6 points against 13.4 for Mistral, and the specialists continue to hedge on 47–63% of neutral questions (baseline 91%). The prompt-aware protocol attributes most of the remaining central scores to both-sides answers (hedge flags on 53–

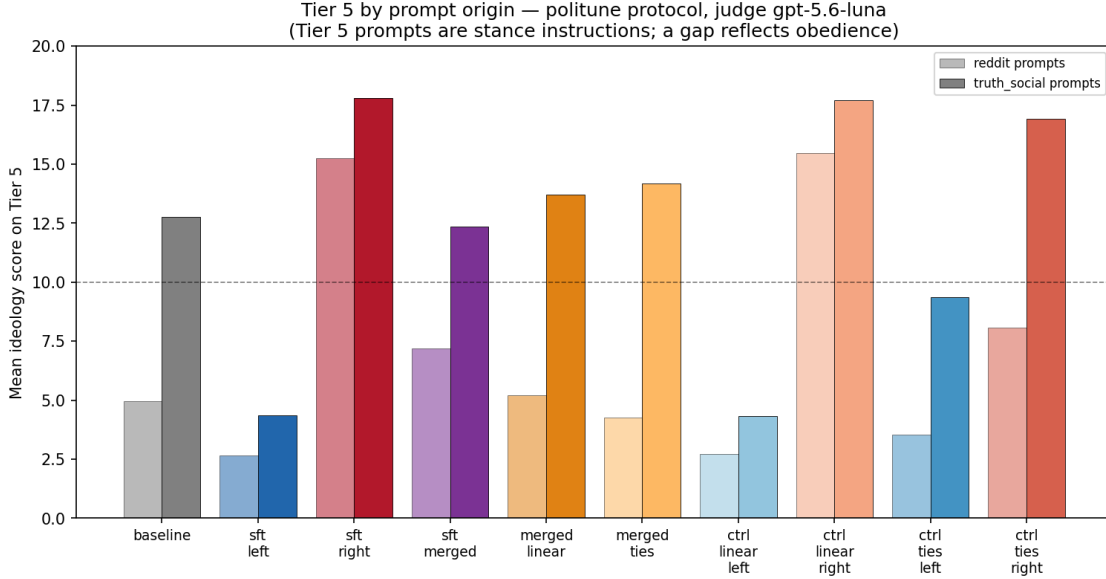


Figure 4: Tier 5 (held-out PoliTune stance instructions) by prompt origin. The gap between Reddit-origin and Truth Social-origin instructions measures how far a model follows the instruction’s stance.

67% of specialist answers, unscorable 13%), so the base model’s default of presenting both sides is reduced but not overridden. The adapter geometry is consistent with this: the left and right deltas of the same seed have cosine similarity 0.35 (Mistral 0.10) and two seeds of the same specialist 0.55–0.57 (Mistral 0.22–0.25), so a larger fraction of each Gemma adapter is common to both ideologies. On the stance instructions every Gemma instance follows the instruction (origin gaps 5.7–10.3 for all ten conditions, against about 2 for the Mistral specialists).

Within the smaller shift, the structural findings hold. The pooled model, SFT-Merged, sits at 9.2 and has the largest within-prompt variance share of any condition (0.46 for both seeds, against 0.25–0.30 for the specialists and 0.10 for the baseline): on the same question it answers from either side on different samples. Its total standard deviation (4.4) is only slightly above the specialists’ (3.8–4.1; Brown–Forsythe $p = 0.10$ against SFT-Left and 0.01 against SFT-Right), because on Gemma 4 the specialists are themselves bimodal, alternating between a committed answer and a hedge. Linear averaging of two seeds of the same specialist is again inert (7.6 and 12.5 against 7.4 and 13.0; $p = 0.60$ and 0.28), and averaging the opposing adapters again yields a central mean (9.4) with a hedge rate between the specialists’ and the baseline’s. TIES merging at density 0.5 removes the adapters’ effect entirely: both same-ideology TIES controls score 9.4 and 9.7 with hedge rates of 91–92% and variances indistinguishable from the untrained baseline ($p < 10^{-3}$ against their specialists), and Merged-TIES is likewise indistinguishable from the baseline (9.4, hedge 0.90). The Pareto partition is the same as for Mistral: the frontier holds the specialists and the baseline, and every merged condition is dominated.

4.9 Judge Robustness

All reported scores come from a single judge. Two checks are available. First, the two protocols agree on every finding above: condition means differ by at most 0.9 points between protocols on the neutral questions, and the ordering, the variance pattern and the Pareto partition are identical. Second, before the second judge was stopped for cost, `gemini-3.8-flash` had scored 3,780 responses

Table 6: Both base models on the 43 neutral questions (PoliTune protocol, judge `gpt-5.6-luna`): mean, standard deviation, hedge rate, within-prompt variance share, and the Tier 5 origin gap. Mistral uses three seeds and 150 Tier 5 prompts; Gemma 4 uses two seeds and 60 Tier 5 prompts. The last row is the mean cosine similarity between the left and right adapter deltas of the same seed.

Condition	Mistral-7B-Instruct-v0.2					Gemma-4-12B-it				
	$\bar{s}$	σ	hedge	within	T5 gap	$\bar{s}$	σ	hedge	within	T5 gap
Baseline	8.5	3.0	0.80	0.05	7.8	9.4	2.2	0.91	0.10	7.4
SFT-Left	3.0	3.1	0.06	0.21	1.7	7.4	4.1	0.63	0.27	7.8
SFT-Right	16.4	2.7	0.06	0.33	2.5	13.0	3.8	0.47	0.28	5.7
SFT-Merged	9.4	6.8	0.08	0.42	5.1	9.2	4.4	0.64	0.46	7.3
Merged-Linear	7.7	6.4	0.11	0.30	8.5	9.4	3.8	0.76	0.25	10.3
Merged-TIES	6.4	5.3	0.22	0.22	9.9	9.4	2.5	0.90	0.12	9.8
Ctrl-Linear (L+L)	3.0	3.0	0.07	0.18	1.6	7.6	3.8	0.70	0.21	7.6
Ctrl-Linear (R+R)	16.6	2.6	0.01	0.36	2.3	12.5	4.0	0.55	0.32	5.7
Ctrl-TIES (L+L)	3.9	3.8	0.18	0.28	5.8	9.4	2.3	0.91	0.11	8.0
Ctrl-TIES (R+R)	13.5	4.8	0.18	0.21	8.8	9.7	2.3	0.92	0.29	8.8
Left/right adapter cosine			0.10			0.35				

under the PoliTune protocol (baseline, SFT-Left, SFT-Merged, Merged-Linear and a control). On that subset the two judges have Pearson $r = 0.93$, Krippendorff’s $\alpha = 0.91$, Cohen’s $\kappa = 0.61$ on five ordinal bins, and a mean absolute difference of 1.3 points. This is a partial check and is listed as a limitation.

4.10 Hypothesis Assessment

H1 (SFT shifts ideology): supported. Symmetric shifts of about six to eight points, reproduced across three seeds with instance means within 0.3 points.

H2 (merged models reproduce the mixture): supported, with a qualification. SFT-Merged is bimodal with twice the specialists’ variance, 42% of it within prompts. The qualification is that the other 58% is between prompts: on some questions the mixture resolves to a stable position, so the model samples from the mixture on some prompts and holds a fixed position on others.

H3 (weight-space merging fails to produce moderation): supported for linear averaging; confounded. Linear controls show that averaging alone does not change the model, so Merged-Linear’s variance is attributable to the opposing adapters. The TIES control degrades a same-ideology merge, so Merged-TIES’s lower variance, leftward mean and higher hedge rate cannot be separated from merge damage.

H4 (merged models do not hedge more than the baseline): supported. Hedge rates of 8–22% for merged conditions against 80% for the baseline; the merged conditions’ central means come from opposing committed answers.

Gemma-4-12B-it. H1 holds with a smaller effect (5.6 points between specialists), H2 holds for the within-prompt share (0.46 against 0.25–0.30) but not for the doubling of total variance, H3 holds for linear averaging and TIES is again confounded, and H4 holds in the weak sense that no merged condition hedges more than the baseline, although all Gemma conditions hedge on at least 47% of questions.

5 Discussion

5.1 What Aggregation Produced

Fine-tuning on pooled opposing data produced a model that reproduces the pool. This is the expected maximum-likelihood outcome. This study measures that outcome at the level of individual answers: bimodal scores, a large sampling component, a prompt-dependent mixing weight, no increase in hedging, and no loss of fluency. Each answer is fluent and committed; which side it commits to is partly a draw from the training mixture.

Weight-space merging of separately trained specialists produces the same outcome through a different mechanism. The two deltas are nearly orthogonal, so averaging does not cancel them; it superimposes two shifts of half magnitude, and the merged model exhibits both. The linear controls show that the averaging operation itself does not cause this. Merged-TIES has the lowest variance of the merged conditions, but the same-ideology control shows that TIES degrades adapters and increases hedging, which lowers variance independently of any conflict resolution.

5.2 Hedging, Moderation and the Baseline

A mean near 10 could be read as moderation. The prompt-aware protocol shows that the only model in this study that is near 10 and consistent, the untrained baseline, declines to take a position on 80–86% of neutral questions. None of the trained models hedges at that rate; the merged models reach the centre by averaging committed answers on both sides. No model in this experiment commits to centrist positions consistently, although the design did not preclude one. Whether other aggregation strategies (conditioning on group, mixture-of-experts routing, group-aware objectives such as GPO) can produce one remains open.

5.3 Dependence on the Base Model

The Gemma 4 replication separates two things the Mistral results conflate: how far a fixed adapter recipe can move a base model’s ideology, and what aggregation does to whatever movement exists. The first depends on the base model. Rank-16 adapters on the attention projections, trained for two epochs on about 2,500 examples, move Mistral-7B-Instruct-v0.2 by 13 points between specialists and Gemma-4-12B-it by 6, and the Gemma specialists keep the base model’s both-sides default on half of the questions. The higher similarity between Gemma’s left and right adapters indicates that much of what the adapters learn on this model is the response format shared by both datasets rather than ideology. The second does not depend on the base model in our data: on both, pooled training produces a model that samples between the sides within a prompt, same-ideology averaging is inert, opposing-adapter averaging is central, and TIES merging at density 0.5 damages the adapters, on Gemma 4 to the point of erasing them. A larger adapter budget or full fine-tuning would likely move Gemma 4 further; the reduced replication does not establish how the two effects scale with the strength of the manipulation.

5.4 Relation to Arrow’s Theorem

Arrow’s theorem is about aggregating ordinal preference profiles over three or more alternatives under unanimity, independence of irrelevant alternatives and non-dictatorship. Our experiment measures none of these conditions and aggregates responses rather than preference orderings. The connection is interpretive: pooled fine-tuning acts like a per-example random dictator (each training example is one voter’s answer), and the resulting policy is a lottery over the voters, which is what

a random-dictator rule produces. Positional-voting readings of preference learning [Siththaranjan et al., 2024] concern pairwise preference objectives that this study does not use. The results are therefore consistent with the social-choice expectation that pooling opposed preferences without recording who holds them does not yield a coherent collective position. They are not a test of the theorem.

5.5 Limitations

Two base models, one adapter recipe. The main results are for Mistral-7B-Instruct-v0.2; the Gemma-4-12B-it replication uses two seeds and a smaller Tier 5 set, and the recipe moves that model much less. Rank-16 QLoRA on the attention projections is the only training configuration tested; full fine-tuning or other adapter targets could change both the size of the shift and how the mixture is represented.

Single judge. Scores come from `gpt-5.6-luna`; the Gemini check covers a fifth of the records and one protocol. LLM judges share biases, and a human-rated subset was not collected.

Synthetic, stance-laden training data. PoliTune’s prompts are instructions with a built-in stance and its responses are synthetic; the dataset cards do not name the generator. The neutral-question set is small (43 prompts) and hand-written.

One aggregation ratio, one adversarial construction. The merged data is a 50/50 pool with targets assigned at random. Weighted pools, group-conditioned training and routing were not tested, so the results bound the naive strategy only.

TIES with one density. Only density 0.5 was used, and the control shows it damages adapters; other densities or DARE-style merges might behave differently.

No direct Arrow test. As discussed, no Arrow condition is measured; the social-choice framing is interpretive.

5.6 Implications

For alignment pipelines that pool annotators with opposing values, pooled SFT produces a model that samples from the pooled values, and the model’s consistency depends on the prompt as well as on the model. Evaluations that score one sample per prompt, or that use stance-laden prompts, will not detect this. Two inexpensive measurements detect it: repeated sampling with a within-prompt variance decomposition, and a judge that can report “no position”. For weight-space merging, same-ideology control merges should accompany any cross-ideology result, since the merge operation can itself move a model towards the centre.

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